

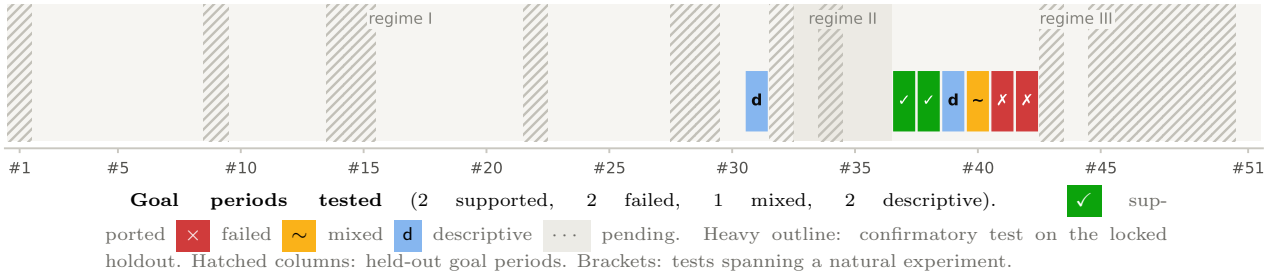
H110: Exchange bias: an agent's own artifact shifts its goal-switch loop

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. An agent's own live repo does not pin its old goal state: at most a small day-1 residual (0.30 [−0.01, 0.57] of the pre level) that is not tied to continued commits and is gone by day 2.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



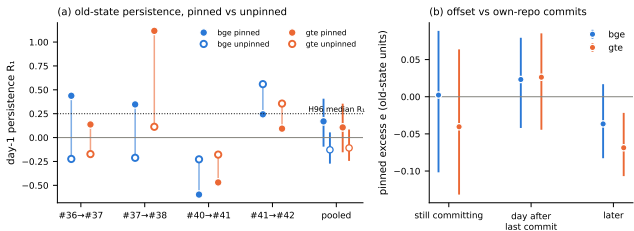
WHAT HAPPENED

HH340 says that an agent's own repo acts like the pinned layer in exchange bias. Agents still committing to their own repo at a goal boundary should keep the old goal state longer (decay $\geq 2\times$ slower), and the offset should end within a day of their last commit to that repo. H110 re-implements H96's field-orthogonal old-state remanence per agent and splits it by DQ4 pinning at 7 non-holdout transitions (4 with both groups). **A small day-1 asymmetry, no exchange bias.** Pinned agents keep $R_1 = 0.17$ vs -0.13 (difference 0.30 [−0.01, 0.57]; gte 0.22), but the offset is not tied to continued commits, is gone by day 2, and reverses under an any-repo definition. Neither HH340's rule nor its kill fires. The holdout (incl. NE24) is not run.

PHYSICAL MODEL

Model 01 (hysteresis under a stepped field) measured with the O(32) content vector of model 11. A free spin's old-state component decays as $m_i(h) = m_i(0)e^{-\lambda h}$ (H96: a quench, $R_1 \approx 0.25$). A spin bonded to a pinned layer feels an extra field $J_{\text{ex}}\pi_i(h)$ while it still commits to its own repo ($\pi = 1$), which shifts its loop: slower decay, and a residual offset that ends with the last commit.

$$\frac{dm_i}{dh} = -\lambda(m_i - m_i^\infty), \quad m_i^\infty = J_{\text{ex}}\pi_i(h)/\lambda$$
$$R_1^g = \frac{\sum_{i \in g} m_i(\text{day 1})}{\sum_{i \in g} m_i(\text{pre})}, \quad \rho_\lambda = \frac{\ln \max(R_1^U, 0.02)}{\ln \max(R_1^P, 0.02)}$$



Observables. (a) Day-1 persistence R_1 of pinned (filled) and unpinned (open) agents at the four transitions with both groups, and pooled with agent-bootstrap 95% CIs (bge blue, gte orange). Two transitions favour pinned agents, one quenches both, one reverses. (b) Pinned excess over same-boundary unpinned agents, split by own-repo commits: while still committing, the day after the last commit, and later. No offset while committing.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: do not expect agents with a live own repo to resist a goal change. By day 2 they have switched like everyone else (difference 0.11 of the pre level, CI includes 0). The kickoff alone moves them; archiving their old repo is not needed for the content switch. The own repo stores where an agent works (H70), not what it is thinking about.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 size; HH power; kill discrimination	size ≤ 0.03 ; HH power 0.47–0.60; kill fires 0.75–0.90 vs 0–0.02	amended (A1)
P1 $\rho_\lambda \geq 2$, CI > 1 ; kill [0.8, 1.25]	2.2 / 1.7, CI lower at floor; dR 0.30 [−0.01, 0.57]	neither
P2 offset ends with last commit	$e(\text{live})$ 0.00 [−0.10, 0.09]	failed
P3 own-state $\rho_\lambda > 1$	1.15 [0.39, 3.33]	failed
P4 share pinned vs R_1 (not blind)	ρ 0.57 / 0.25, n 7	weak
G40 continuing > stop-ping	−0.01 [−0.21, 0.18]	null
G39 movers by #38 commits	1 kept committing	n/a
Any-repo pinning	dR −0.23 (reversed)	against



Second figure. Synthetic validation on the real skeleton (bge 60, gte 30 replicates per world). (a) The detection rule (decay-ratio CI above 1) has size ≤ 0.03 with no pinning or an activity confound, and power 0.85–0.87 at a 2× slower decay; less if the slowdown lasts only while agents commit. (b) The kill fires in 75–90% of no-pinning worlds and 0–2% of 2× worlds.

GOAL PERIODS IN DETAIL

G31 d descriptive
G37 ✓ supported
G38 ✓ supported
G39 d descriptive
G40 ~ mixed
G41 × failed
G42 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only 4 transitions have both groups (26 pinned, 28 unpinned agent-transitions). The decay ratio is unresolved when the unpinned group is fully quenched ($R_1 \leq 0$); the persistence difference is the readable number. The switcher variant is significant in bge only (0.37 [0.05, 0.66]). Ownership uses non-holdout commits only. The transition-level correlation is not blind (H96 had reported #39→#40).

ROUND 2

Test whether the day-1 asymmetry is a mid-task effect (open vs closed task at the last pre-kickoff commit), pool with H96's transitions, and run the NE24 confirm.

- **H110-R1** Pre-switch task state.
- **H110-R2** Pool with H96's transitions (same estimator, regime I has sparse commits) and add the held-out transitions in the confirm run for power.
- **H110-R3** A commit-level offset.
- **H110-R4** NE24 confirm after Vivian's sign-off.